

Sai Keerthan Kasula

Cryptography & PKI Engineer | Post-Quantum Cryptography | HSM / PKCS#11 | Cryptographic Modernization

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Professional Summary

Security Engineer with 3.5 years of experience securing India's national banking infrastructure (**RBI SFMS – 238 banks, 2–3 crore transactions/day**). Specialized in enterprise PKI, HSM/PKCS#11 integration, cryptographic modernization, X.509 certificate lifecycle management, and cryptographic governance. Hands-on experience benchmarking post-quantum cryptographic algorithms (ML-DSA, ML-KEM) against RSA/ECC baselines at production-scale transaction volumes. Strong background in Java-based secure backend engineering, cryptographic performance optimization, and financial security infrastructure.

Technical Stack

- **Cryptography:** AES-GCM, RSA, ECC, ML-DSA, ML-KEM, SLH-DSA (SPHINCS+), Digital Signatures, Hybrid Cryptography, Bouncy Castle API, OpenSSL
- **PKI & Certificate Management:** X.509, CRL, OCSP, ASN.1, RFC/PKCS Standards, Certificate Lifecycle Management, Crypto-Agility, CBOM Inventory
- **HSM & Key Management:** IBM HSM, PKCS#11 Integration, HSM-Backed Signing, Multi-Vendor HSM Support, Secure Token Management
- **Security & Compliance:** FIPS 140-3, FIPS 203/204, NIST IR 8547, CNSA 2.0, STRIDE Threat Modeling, Vulnerability Remediation, Audit Readiness
- **DevOps & Cloud:** Docker, GitHub Actions, AWS EC2, SonarQube, Burp Suite
- **Languages & Frameworks:** Java, Spring Boot, Maven, SQL

Professional Experience

Tata Consultancy Services (TCS) | **Client: IFTAS, Subsidiary of Reserve Bank of India** Jan 2023 – Present
Security Engineer – SFMS (Structured Financial Messaging Service)

- Established CBOM-aligned cryptographic asset governance across **230+ banking nodes**, enabling enterprise-wide PQC readiness assessment and RBI audit compliance through centralized certificate and crypto metadata inventory management.
- Architected and delivered a zero-downtime **AES-256-GCM migration** strategy across the SFMS ecosystem; phased backward-compatible rollout reduced deployment complexity by **50%** compared to simultaneous migration approaches.
- Led ecosystem-wide **algorithm deprecation** and ASN.1/RFC-compliant digital signature governance across **60+ banks**; successfully drove migration away from **MD5** and **SHA-1** within a 7-month timeline.
- Reduced **SFMS Signer** signing latency by **50%** (200 ms → 100–120 ms); built a custom PKCS#11 integration layer with automatic HSM slot detection and certificate metadata caching for enterprise signing workflows.
- Modernized Java-based cryptographic signing infrastructure during **JDK 8 to JDK 17** migration by resolving strong-encapsulation compatibility issues in PKCS#11 and HSM integrations.
- Initiated and led internal **PQC readiness benchmarking**; evaluated ML-DSA and ML-KEM against RSA/ECC baselines at production-scale financial transaction volumes and presented migration roadmap findings to **TCS NGM India leadership**.
- Supported enterprise CA operations including **CRL validation**, trust-chain verification, HSM-backed signing token operations, and certificate authority configuration across India's national interbank settlement infrastructure.
- Built CI/CD pipelines for internal PKI tooling with integrated **SAST**, Docker containerization, and **AWS EC2** deployment workflows for secure cryptographic operations.

Projects

Secure Hybrid Crypto Engine

Developed a full-stack cryptographic platform for exploring and benchmarking classical and post-quantum cryptographic algorithms through a unified interface. Built using Java, Spring Boot, hybrid cryptographic workflows, and PQC libraries to support secure experimentation, algorithm comparison, and enterprise cryptography research workflows.

GitHub: github.com/sai-keerthan/secure-hybrid-crypto-engine

Education

Chaitanya Bharati Institute of Technology

Jun 2020 – Aug 2022

Master of Computer Applications (MCA) | CGPA: 8.55 / 10.0

Awards & Publications

- **TCS Beyond Excellence Award** – recognized for strengthening cryptographic governance across RBI's national banking ecosystem through ASN.1/RFC-compliant digital signature standardization.
- **Academic Excellence Award** – awarded for outstanding academic performance throughout the MCA program.
- Technical author on cryptography, PKI architecture, and secure backend engineering: **keerthankasula.hashnode.dev**